

# Kasper Rømer Grøntved

AI Engineer · Colourbox



AI engineer working on machine learning systems for commercial search, hosted on European infrastructure. PhD in multi-robot systems from the University of Southern Denmark, with a research stay at Carnegie Mellon's Robotics Institute.

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## EXPERIENCE

Jun 2026 – now

### AI Engineer · Colourbox

Building the machine learning systems behind Colourbox's commercial search products, hosted end to end on European infrastructure and kept within EU data residency.

Search that understands the actual content of images, video, audio and documents — moving beyond keywords toward genuine semantic and multimodal understanding. The models powering it run on self-hosted, EU-based infrastructure, so customer data and AI processing never leave European jurisdiction. Favouring sovereign, right-sized architecture over unnecessary complexity or reliance on non-EU providers.

Sep 2025 – Jun 2026

### MLOps · Esoft

In-house machine learning models and pipelines for real estate image processing at production scale — classification, segmentation, object detection and generative models running on GPU clusters under real SLAs, across thousands of orders a day.

May 2022 – Aug 2025

### PhD Researcher · SDU UAS Center, University of Southern Denmark

PhD on human–multi-robot interaction as part of the HERD project — Human-AI Collaboration: Engaging and Controlling Swarms of Robots and Drones — funded by Innovation Fund Denmark.

Researched how a single operator can engage with and control swarms of autonomous robots and drones in unstructured, outdoor environments, with real-world use cases alongside industry partners in agriculture and search & rescue. Contributed decentralized trajectory task allocation, coverage path planning, single-console swarm control, and UAV-to-UAV communication, all validated in outdoor field trials. Position granted out of 42 applicants. Six peer-reviewed publications; two M.Sc. theses supervised.

Sep 2024 – Dec 2024

### Visiting Scholar · Carnegie Mellon University Robotics Institute

Research stay in Pittsburgh, Pennsylvania, during the PhD.

Jun 2020 – May 2022

### Systems Engineer · a2i systems A/S

A self-learning pricing platform used by fuel, EV-charging and convenience-retail sites internationally. Worked across machine learning and operations to keep a real-time, data-driven AI product running reliably in production, supporting the models that forecast customer behaviour and optimize pricing in dynamic markets — early grounding in taking AI from prototype to dependable, large-scale operation.

Nov 2018 – Jun 2020

### Student Programmer · a2i systems A/S

Alongside the B.Sc. and M.Sc. programmes. Linux system administration and Git.

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## EDUCATION

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|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 – 2025 | <b>PhD · University of Southern Denmark</b><br><i>Cooperative Control of Multirobot Systems in Real-World Applications.</i> Defended 2025.<br>Supervisors: Anders Lyhne Christensen, Ulrik Pagh Schultz Lundquist. |
| 2018 – 2022 | <b>M.Sc. Advanced Robotic Systems · University of Southern Denmark</b><br><i>Multi-Agent Decentralised Coordination using CNRL for Industrial Applications.</i>                                                    |
| 2015 – 2018 | <b>B.Sc. Robotic Systems · University of Southern Denmark</b><br><i>Semantic Segmentation using a Deep Neural Network for Pose Estimation of a Rigid Object.</i>                                                   |

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## SKILLS

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| LANGUAGES      | Python · C / C++ · Rust · TypeScript · Bash · Java · SQL · LaTeX                                                                                                                                             |
| AI & ML        | Multimodal and semantic search · embeddings and retrieval · LLM agent frameworks · vLLM serving · PyTorch · computer vision (classification, segmentation, detection) · generative models · model evaluation |
| MLOPS          | Self-hosted GPU clusters · model serving under SLA · data pipelines and versioning · EU-resident infrastructure and GDPR-constrained architecture                                                            |
| ROBOTICS       | ROS / ROS 2 · multi-agent task allocation · coverage path planning · FreeRTOS · SITL & HITL simulation · geospatial processing                                                                               |
| INFRASTRUCTURE | Docker · Unix systems and Linux administration · Git · CI/CD · data engineering                                                                                                                              |
| SPOKEN         | Danish (native) · English (fluent — reading, writing, speaking)                                                                                                                                              |
| CERTIFICATIONS | Certified A1/A3 drone pilot · Drone Pilot Assistant                                                                                                                                                          |

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## RESEARCH OUTPUT

[grontved.xyz/publications.html](https://grontved.xyz/publications.html)

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| JOURNAL & CHAPTERS | Drones 9(9):628, 2025 (SAREnv) · book chapter in Cultural Robotics, Springer, 2023                                                                                                                                      |
| CONFERENCES        | ICUAS 2025 · DCAI 2024 · ICUAS 2024 (×2) · ICAR 2023 · IROS 2022 (adjunct)                                                                                                                                              |
| FUNDED PROJECTS    | HERD — Human-AI Collaboration: Engaging and Controlling Swarms of Robots and Drones · DIREC (Innovation Fund Denmark) · NAMUR (Independent Research Fund Denmark) · WildDrone MSCA Doctoral Network (EU Horizon Europe) |
| SUPERVISION        | Three M.Sc. students across two theses, both successfully defended in 2023                                                                                                                                              |
| OPEN SOURCE        | TrajAllocPy · TrajGenPy · SwarmTalk · DDICP · SAREnv · HopDatabase                                                                                                                                                      |